Credit Instruments and Risk Homework (#7)

(Courtesy of Prof. Leif Andersen for the problems and the material)

Exercise 1

Consider a corporate bond:

- with notional N (compare to \$1 in the lecture notes);
- issued by a firm with constant default intensity λ (not a function of time as in the notes)
- final maturity T, at which the notional is returned to the bond holders;
- constant continuous coupon c, i.e. in any time interval $[t, t+dt]$, where $t < T$, the bond holders are paid $N \cdot c \cdot dt$ as long as there is no default prior to t;
- recovery rate of R, i.e. at the time of default, if it happens before T, the bond holders get paid $N \cdot R$, where R is between 0 and 1;

And we further simplify, by assuming that the risk-free interest rate is constant, flat at $r > 0$.

Case 1, $R = c = 0$, i.e. bond is zero coupon and has no recovery.

a) What is the bond value at $t = 0$?

Case 2, $R = 0$, but $c \neq 0$, i.e. bond has no recovery but pays coupons.

a) What is the value of the coupon stream at $t = 0$?
b) What is the total value of the bond at $t = 0$?
c) What is the limit of the initial bond value as its maturity, $T \to \infty$**?**
d) At which value of c is the initial bond value = N? (par coupon)

Case 3, $R \neq 0$, and $c \neq 0$, i.e. bond has recovery and pays coupons.

a) What is the value of the recovery component at $t = 0$?
b) What is the total value of the bond at $t = 0$?
c) What is the limit of the initial bond value as its maturity, $T \to \infty$**?**
d) At which value of c is the initial bond value = N? (par coupon)

Exercise 2

Consider a CDS contract. Recall from the lecture notes, that its par spread is the coupon rate that makes the value of the CDS be perfect zero at initiation.

$$c_{par} = \frac{C(0; 1 - R)}{\sum_{i=1}^{N} \delta_i B(0, T_i)}$$

When λ is roughly constant in time, this par spread can be well approximated by

$$c_{par} \approx \lambda(1 - R)$$

a) Now, assuming that the CDS is contract on the bond described in exercise 1 (do not mistake the par spread on a CDS from a par spread on a bond !!), show that the above approximation works.
b) Provide an intuition why such a relationship makes sense from credit risk products perspective.

Exercise 3

Consider a situation where only three ratings exist. G (good), B (bad) and D (default). Let's suppose the quarterly credit rating migration is described by a Markov chain (homogeneous in time!) with the following transition matrix:

$$M\left(\frac{1}{4}\right) = \begin{array}{cccc} & G & B & D \\ G & ? & 8\% & 3\% \\ B & ? & 90\% & 4\% \\ D & ? & ? & ? \end{array}$$

a. Infer the missing values (assuming no credit singularities of any kind)
b. What is the transition matrix for a half-year interval? (***M(1/2)***)
c. What is the probability of default in ½ year, if one starts from good rating?